

Abstract

The rapid advancement of Artificial Intelligence has led to the widespread creation of highly realistic synthetic media, commonly known as deepfakes. While these technologies offer creative possibilities, they also pose serious challenges such as misinformation, identity fraud, and loss of digital trust. Existing deepfake detection systems often suffer from limitations such as lack of transparency, poor generalization, and dependence on single-modality analysis.

This project, titled “TRUTH-SEEKER: A Hybrid Deepfake Detection System”, proposes an efficient and reliable solution for detecting manipulated content. The system adopts a hybrid edge-cloud architecture, combining fast, lightweight detection on user devices with detailed and high-accuracy analysis on cloud servers. This approach ensures both low latency and improved detection performance.

The proposed system utilizes multi-modal analysis, integrating image, video, and audio data to identify inconsistencies such as visual artifacts, temporal irregularities, and audio mismatches. Additionally, Explainable Artificial Intelligence (XAI) techniques are incorporated to provide visual and textual explanations, enhancing transparency and user trust.

A key feature of the system is the integration of a fact-checking module, which verifies the authenticity of associated information, extending the functionality beyond simple detection. The system is implemented using standard machine learning frameworks and evaluated on benchmark datasets and real-world samples.

The results demonstrate that the proposed system achieves improved accuracy, better generalization, and enhanced interpretability compared to traditional methods. Overall, TRUTH-SEEKER provides a scalable, efficient, and user-friendly solution for deepfake detection and verification in real-world applications.

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Chapter 1

Introduction

1.1 Background of the project

The rapid advancement of Artificial Intelligence (AI) and machine learning in the late 2010s and early 2020s has fundamentally revolutionized the landscape of digital media creation, editing, and distribution. Central to this paradigm shift is the advent of highly sophisticated generative models, which have led to the proliferation of AI-generated or manipulated content, colloquially known as “deepfakes.” The term, a portmanteau of “deep learning” and “fake,” originally emerged in late 2017 to describe synthetic media where a person in an existing image or video is replaced with someone else’s likeness. Since then, the technology has evolved exponentially, transitioning from rudimentary face-swapping algorithms to highly advanced generation frameworks capable of synthesizing photorealistic visuals and hyper-realistic audio from scratch.

This technological leap is primarily powered by several underlying architectures. Generative Adversarial Networks (GANs), introduced in 2014, pioneered the space by employing a dual-network system a generator and a discriminator locked in a continuous zero-sum game, resulting in outputs of startling realism. More recently, Diffusion Models, such as those utilized by Midjourney, Stable Diffusion, and OpenAI’s DALL-E and Sora, have pushed the boundaries further, offering unprecedented control over text-to-image and text-to-video generation. Concurrently, advancements in neural voice cloning and text-to-speech (TTS) synthesis have made it possible to replicate a target’s voice with immense accuracy using only a few seconds of sample

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audio.

While these generative technologies possess immense potential for legitimate applications such as personalized education, post-production in the film industry, synthetic data generation for medical research, and enhanced accessibility tools—they inherently present severe ethical, security, and societal risks. The democratization of these tools has lowered the barrier to entry, allowing individuals without specialized technical expertise to generate highly convincing synthetic media. Consequently, deepfakes are increasingly being weaponized. Malicious actors leverage these tools for a variety of illicit purposes, including large-scale misinformation campaigns, political manipulation, financial fraud through social engineering, identity theft, and the creation of non-consensual explicit material.

The widespread dissemination of malicious deepfakes threatens to erode public trust in digital media, creating a phenomenon often described as the “liar’s dividend,” where the mere existence of deepfake technology allows individuals to dismiss genuine evidence as artificially generated. In response to this growing threat, the scientific community has engaged in a continuous arms race to develop robust deepfake detection mechanisms. Early detection systems relied on identifying visual artifacts, such as irregular blinking patterns, unnatural facial boundaries, or spatial inconsistencies. However, as generative models improve, these discernible artifacts are rapidly disappearing, rendering traditional heuristic-based detection methods obsolete. This dynamic necessitates the development of sophisticated, multi-modal, and adaptable detection frameworks capable of analyzing complex digital media at a granular level, thereby restoring integrity to the digital information ecosystem.

1.2 Motivation

The primary motivation for this project stems from the critical and persistent limitations inherent in current state-of-the-art deepfake detection systems. While the academic community and industry leaders have developed models that achieve high classification accuracy in controlled laboratory environments, these existing solutions face significant functional and operational challenges that hinder their efficacy and reliability when deployed in real-world scenarios.

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First and foremost is the pervasive “Black-Box” Problem. The majority of modern deepfake detectors rely on deep neural networks that output binary classifications simply labeling media as “Real” or “Fake”—or providing a raw probability score. They do not offer any underlying rationale or contextual evidence to explain why a specific decision was reached. In high-stakes environments, such as journalism, legal proceedings, or content moderation for social media platforms, a simple binary output is insufficient. Without transparency, it is nearly impossible for human analysts to trust the system’s output. Therefore, there is a pressing need to integrate Explainable AI (XAI) paradigms to demystify these decisions and highlight the specific spatial or temporal anomalies that triggered the detection.

Secondly, current systems suffer from severe Over-Specialization and Poor Generalization. Many detection models are trained on highly specific datasets containing deepfakes generated by a limited set of known algorithms (e.g., specific GAN architectures). Consequently, these models suffer from domain shift; their performance degrades dramatically when they encounter media manipulated by novel, unseen generation techniques, or when analyzing diverse content forms like 3D animations and CGI. A robust system must be designed with generalization in mind, capable of adapting to the rapidly evolving landscape of generative AI.

Thirdly, the industry currently operates in Modality Silos. Existing detectors typically specialize in a single modality—analyzing only the visual frames of a video or exclusively examining an audio track. However, modern deepfakes often encompass both visual and auditory elements. Single-modality systems fail to detect cross modal inconsistencies, such as desynchronization between lip movements and spoken phonemes, or semantic mismatches between the emotional tone of a voice and the corresponding facial expressions. An integrated, multi-modal approach is critical to exploit these inter-modality discrepancies, which remain a significant hurdle for current generation models.

Furthermore, Privacy and Data Security present a massive roadblock to public adoption. The majority of highly accurate deepfake detection tools are hosted on centralized cloud servers due to the massive computational overhead required by deep learning inference. This architecture requires users to upload potentially sensitive, private, or proprietary media to third-party servers, raising severe privacy concerns.

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There is an urgent need for a privacy-preserving framework that shifts the initial burden of computation to local edge devices, allowing users to verify media without compromising their data.

Finally, there is a distinct gap regarding Information Integrity and Contextual Verification. Determining whether an image or video is synthetically generated solves only half of the modern misinformation puzzle. Even authentic media can be repurposed and shared with false context. Current systems do not bridge the gap between media forensics (detecting digital manipulation) and fact-checking (verifying the semantic claims associated with the media). This project is motivated by the vision of a holistic platform that not only detects AI manipulation but also cross-references the contextual claims against real-world data, providing users with a comprehensive analysis of digital truth.

1.3 Scope

This project focuses on the comprehensive design, development, optimization, and rigorous evaluation of a hybrid, multi-modal AI system. This system is specifically engineered to detect AI-generated media, explain its reasoning, and verify the contextual factual information associated with the media. To ensure the project remains focused and achievable within the designated timeframe, the core deliverables and explicit boundaries have been unified as follows:

- **Multi-modal Detection Architecture:** The development and integration of distinct models for visual analysis (detecting spatial anomalies, blending errors, and GAN/diffusion artifacts) and audio analysis (detecting synthetic voice signatures and frequency anomalies). The system will analyze these modalities both independently and synchronously.
- **Integration of Explainable AI (XAI):** The implementation of state-of-the-art interpretability techniques, utilizing algorithms like LIME and SHAP to generate visual heatmaps that highlight manipulated regions, alongside Chain-of-Thought (CoT) prompting for natural language explanations of the model's reasoning. This dual approach effectively mitigates the traditional "black

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box" nature of deep learning, ensuring that every classification is backed by transparent, auditable evidence.

- **Hybrid Edge-Cloud Processing Pipeline:** The design of a dual-stage workflow deploying lightweight, compressed detection models for fast, privacy-preserving inference on edge devices, coupled with an optional, user-consented cloud handoff for deep, high-accuracy multi-modal analysis.
- **Automated Fact Verification Component:** The development of a module leveraging real-time web-search integration and Large Language Models (LLMs) to verify factual claims associated with uploaded media, addressing contextual misinformation.
- **Benchmarking and Comparative Evaluation:** A rigorous evaluation of the proposed system against state-of-the-art baselines (e.g., AVT2-DWF, LIPINC V2, and SpecTTTra) on standardized deepfake datasets (e.g., FaceForensics++, DFDC) to measure metrics such as Accuracy, F1-Score, AUC-ROC, and inference latency.
- **Defensive Prototype Constraint:** The project is strictly defensive and academic in nature. It explicitly excludes the creation or training of new deepfake generative models. Furthermore, the deliverable will be an academic proof-of-concept, excluding commercial software packaging, app store deployment, or direct API integration into major social media platforms.
- **Multimedia Trigger Requirement:** The system focuses on media-driven context and will not serve as a general-purpose, text-only fake news detector; the primary trigger for the system must be an audio, video, or image file.
- **Inference Optimization Focus:** While the project addresses inference efficiency via edge computing, it does not attempt to solve the fundamental computational constraints or energy costs associated with the initial training of massive deep learning models on cloud infrastructure.
- **Interactive Visualization Dashboard:** The development of a user-centric web interface designed to display detection verdicts alongside the XAI outputs

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(e.g., visual heatmaps and CoT reasoning). This dashboard will serve as the primary interaction layer, translating complex multi-modal model outputs into actionable, easily interpretable insights for non-expert users.

- **Data Privacy and Media Retention Limits:** The establishment of strict data lifecycle protocols to address privacy concerns. Media analyzed via the edge pipeline will be processed entirely in-memory with zero local retention, while media escalated to the cloud pipeline will be subjected to ephemeral processing and immediately purged post-inference.
- **Adversarial Robustness Evaluation:** The inclusion of targeted stress-testing to evaluate the multi-modal architecture against common evasion techniques and real-world degradations (e.g., adversarial noise injection, social media compression artifacts, and resolution downscaling) to establish the system's operational resilience.

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Chapter 2

Literature Review

The paper [1] proposes DAVID-XR1, a novel approach for detecting AI-generated videos using explainable visual-language reasoning. The authors build a dedicated dataset called DAVID-X containing 747 videos, including 416 AI-generated and 331 real samples from 15 generators. Their method integrates a vision-language multimodal model with chain-of-thought (CoT) fine-tuning to enable both classification and natural language explanations. The model leverages spatio-temporal defect annotations and distills CoT supervision from Gemini 2.5 Pro, followed by supervised fine-tuning that supports spatial and temporal localization of manipulations. Experimental results show that DAVID-XR1 surpasses GPT-4.1, Gemini-2.5, and Qwen2.5-VL-7B on benchmarks such as Veo2, Pika, and Kling, achieving 76.7% detection accuracy and 54.67% explanation precision. The framework demonstrates strong generalization to unseen generators and produces interpretable outputs, though it remains limited by dataset size and reliance on a specific annotation format.

The study [2] introduces BusterX, a multimodal large language model-based framework designed for AI-generated video forgery detection with natural language explanation. The authors construct GenBuster-200K, a large-scale dataset containing 100K real and 100K synthetic videos generated from both open-source and proprietary models. Instead of treating detection as a binary classification problem, BusterX frames it as a visual reasoning task. The system employs Qwen2.5-VL-7B with LoRA fine-tuning and reinforcement learning using a DAPO reward strategy to balance interpretability and detection accuracy. The training pipeline includes supervised

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fine-tuning (SFT) followed by reinforcement learning with chain-of-thought (CoT) prompting. Experimental results show 85.5% accuracy on GenBuster-200K and a 12% improvement over state-of-the-art models on FakeAVCeleb. The model demonstrates robustness against perturbations such as Gaussian noise, JPEG compression, and low frame rates, while explaining its decisions step-by-step, though it requires high computational resources and relies solely on visual modalities.

The work [3] proposes DeCoF, a detector for AI-generated video that learns temporal artifacts through frame consistency rather than focusing on spatial pixel cues. The authors highlight that earlier spatially biased models struggle with unseen generators. DeCoF addresses this by removing spatial dependencies via CLIP feature extraction for each frame and feeding the embeddings into a lightweight transformer with positional encoding to measure inter-frame coherence. The GVF dataset, consisting of 964 real-fake video pairs curated from MSVD and MSR-VTT prompts, is used for training and evaluation. Results show that DeCoF outperforms spatiotemporal neural networks such as I3D, SlowFast, and MViT on unseen generators like Sora, Veo, and Kling. The approach generalizes well, capturing transferable temporal inconsistencies efficiently with lower

computational cost. However, its reliance on CLIP backbones and lack of spatial forgery analysis may limit performance when temporal artifacts are minimal.

The paper [4] presents a transformer-based universal synthetic video detector aimed at identifying face or background manipulations as well as fully AI-generated content. The model utilizes SigLIP-So400M visual embeddings and introduces an Attention Diversity (AD) loss to prevent attention collapse on facial regions, enabling effective detection across varied domains. The training spans multiple datasets, including FaceForensics++, GTA-V, CelebDF, DeepfakeTIMIT, AVID, HifiFace, DeMamba, NYTimes Quiz, and more. The model samples 64 frames per video and applies positional encodings, with classification performed through a multi-head self-attention transformer. Extensive ablation studies examine the effects of frame count, transformer depth, and loss functions. The model reaches near-perfect accuracy on AVID, DeMamba, and GTA-V when trained with cross-entropy and AD loss, and 80% success on the NYTimes in-the-wild challenge. While it generalizes across modalities

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and manipulation types without requiring facial regions, it depends on high-quality embeddings and significant computational resources.

The research [5] explores fake video detection using a Vision Transformer-based framework designed for robustness under compression and few-shot settings. The authors introduce VideoDiffusion, a dataset of over 10,000 diffusion model-generated videos sourced from SEINE, DynamiCrafter, RAVE, TokenFlow, and Text2Video Zero, alongside proprietary generators such as Sora and RunwayML. The approach adopts a frozen ViT backbone pretrained on video captioning to extract high-level semantic and temporal representations. A lightweight three-layer classification head is trained on these features, supported by learned prototypes for open-set and few-shot classification. The model is evaluated against baselines such as CLIP-D, ResNet-50, ResNet3D, AIGVDet, and MISLNet. Results show strong generalization to unseen generators and superior performance under H.264 compression, achieving notable accuracy even with only 10 training samples. Despite its resilience, the frozen ViT may miss subtle spatial artifacts and performs slightly worse on temporally interpolated datasets such as SEINE.

The paper [6] introduces a forensic-oriented augmentation (FOA) strategy to improve the generalizability of AI-generated video detection models across unseen datasets and manipulation techniques. Rather than relying on fragile low-level noise patterns, the authors perturb superficial artifacts while preserving semantic structure, forcing models to learn more robust forensic cues. A Vision Transformer (ViT) backbone is combined with contrastive learning to align real and fake samples under augmented views. Experiments conducted on FaceForensics++, Celeb-DF-v2, and DeeperForensics-1.0 for training, and cross-evaluated on DFDC and WildDeepfake, demonstrate superior performance to baseline methods. The model consistently achieves state-of-the-art AUC and accuracy on unseen data, emphasizing improved transferability. While the approach is model-agnostic and scalable, it primarily focuses on facial deepfakes and may suppress subtle yet important real-world signals if augmentation parameters are not tuned carefully.

The work [7] argues for mechanistic interpretability (MI) as a holistic strategy to understand the internal functioning of AI systems, particularly deep neural networks.

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Drawing inspiration from empirical methodologies in the life sciences, the authors emphasize coordinated discovery strategies such as pattern recognition, decomposition, localization, and recombination. A case study on InceptionV1, referencing curve detectors analyzed through Distill tools, demonstrates how MI reveals emergent circuits and feature-level representations not explicitly programmed at training time. The authors claim MI enables explanations that generalize to novel scenarios, such as typographic attacks in models like CLIP. Compared to local or global explainability methods, MI provides a more faithful and structured account

of internal computation. However, the approach remains resource-intensive and difficult to automate at scale, limiting widespread adoption for very large AI architectures.

The paper [8] proposes the concept of sparse feature circuits for interpreting and editing causal structures within language models. Using sparse autoencoders (SAEs), the authors extract human-interpretable features that approximate the causal influence of subnetworks more effectively than neuron-level circuits. Attribution patching and integrated gradients are employed to locate influential pathways, while the S HIFT (Sparse Human-Interpretable Feature Trimming) technique enables targeted edits to modify behavior, such as reducing gender bias. Evaluations on subject-verb agreement, bias classification, and large-scale behavior discovery tasks show that fewer than 100 sparse features explain most behaviors, significantly outperforming neuron-based approaches requiring 1500+ units. The approach supports scalable unsupervised interpretation, though training SAEs is computationally demanding and residual variance limits full completeness. Current implementations are restricted to relatively small models like Pythia-70M.

The paper [9] investigates the detection of AI-generated songs by analyzing lyrics extracted from audio using automatic speech recognition. The authors adapt a lyrics dataset from Labrak et al., consisting of 3704 human-composed and 3558 AI-generated lyric samples. Whisper-large-v2 is used to transcribe lyrics directly from audio inputs, and semantic as well as stylistic features are obtained through LLM-based embeddings such as LLM2Vec and BGE-ML-GEMMA. A multilayer perceptron (MLP) classifier is trained to distinguish between AI-generated and human-written lyrics. The study shows that transcription-based detection achieves accuracy comparable to ground

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truth text-based methods, while offering greater robustness to audio perturbations. The model generalizes effectively to previously unseen generators such as Suno and Udio, and supports multilingual detection. However, performance declines for underrepresented genres and languages, and transcription quality may affect recall in noisy conditions.

The study [10] introduces SONICS, a comprehensive dataset and benchmarking framework for detecting synthetic songs, along with a transformer-based model called SpecTTTra. The SONICS dataset contains roughly 97,000 audio tracks spanning various durations and genres, enabling evaluation across both short and long sequences. SpecTTTra uses spectro-temporal tokenization and global attention mechanisms to model long-context information more efficiently than conventional CNN or ViT based architectures. The authors compare performance against ConvNeXt, ViT, and EfficientViT, showing that SpecTTTra achieves superior F1 scores on long-duration clips while reducing memory usage by up to 67%. Additionally, the model outperforms human evaluators in detecting fake songs, particularly for extended audio segments. Despite its efficiency, the approach remains limited to English-language data and exhibits lower accuracy on Udio-generated samples compared to Suno-generated ones, with significant GPU requirements for long-sequence training.

The paper [11] addresses the vulnerability of audio deepfake detection models to common manipulation attacks such as noise addition, volume control, and fading. Using the ASVspoof 2019 LA dataset, the authors evaluate over 71,000 samples under seven real-world perturbation scenarios. They introduce CLAD, a contrastive learning based framework combined with length loss to enhance robustness. The system maps similarly manipulated audio samples close in feature space while separating fake and real speech representations. An encoder coupled with a linear classifier, typically using models like AASIST as a backbone, is used for classification. Experimental results show that baseline state-of-the-art detectors like RawNet2 and Res-TSSDNet experience a significant increase in false acceptance rates under manipulation, whereas CLAD maintains FAR below 1.63% across all conditions. Although effective, the approach is evaluated only on a single dataset and introduces additional pre-training overhead, potentially requiring fine-tuning for different backbones.

The work [12] investigates generalizability challenges in audio deepfake detection and proposes a multi-view feature incorporation strategy. The system combines both handcrafted and deep audio features, including MFCC, CQT, Hubert, XLS-R, and WavLM, and evaluates them on ASVspoof 2019 LA for training and ASVspoof 2021 and In-the-Wild datasets for testing. Detection models based on ResNet18 are trained using cross-entropy loss, while feature fusion and selection are facilitated through Gumbel-max sampling and transformer-based modules. The best fusion approach achieves an equal error rate (EER) of 24.27% on the In-the-Wild dataset, outperforming single-feature baselines. Results demonstrate that deep feature fusion improves out-of-domain robustness compared to handcrafted-only systems. However, some deep models fail to generalize effectively, and increased computational cost and complexity are introduced through the fusion pipeline. Performance in real-world noisy conditions still exhibits degradation.

The paper [13] explores voice anti-spoofing through the VANTI framework, which leverages pretrained self-supervised learning (SSL) models such as HuBERT and WavLM to improve deepfake audio detection across unseen domains. The authors analyze the transferability of different SSL layers and propose selecting the optimal feature hierarchy based on its stability and discriminative power. They build lightweight classifiers on top of narrow-band CNNs to maintain efficiency while preserving robustness. Experiments on ASVspoof 2019 LA, ASVspoof 2021 DF, and In-the-Wild datasets show that middle-layer SSL representations produce the best generalization, achieving lower EER than handcrafted features or spectrogram based models. The framework also demonstrates resilience to replay and TTS/VC attacks. However, performance is sensitive to audio preprocessing choices, and high dimensional embeddings escalate memory needs during large-scale deployment.

The study [14] presents a patchout spectrogram autoencoder approach for detecting synthetic speech, emphasizing efficiency and robustness on real-world data. The model first converts raw waveforms into log-mel spectrograms, then applies masked patch based encoding and reconstruction to learn subtle anomalies associated with TTS and VC-generated audio. Training is conducted on ASVspoof 2019 LA, with cross domain evaluation on the ASVspoof 2021 and In-the-Wild datasets. The patchout

mechanism reduces redundancy and computational load, allowing the autoencoder to generalize better to unseen attacks while maintaining low false positive rates. Results indicate that the method surpasses several convolutional baselines in terms of EER and DET performance. Nonetheless, diminished performance is observed under extreme background noise and low-bitrate compression, and the reconstruction objective may overlook higher-level semantic inconsistencies.

The paper [15] proposes a universal detection model capable of identifying deep fakes across multiple media modalities, including audio, video, and images. Rather than relying on modality-specific architectures, the authors employ a multimodal transformer trained with a unified contrastive loss, enabling feature alignment across domains. The training corpus includes large-scale datasets such as FaceForensics++, DFDC, ASVspoof 2019, and custom synthetic imagery generated by Stable Diffusion and StyleGAN. By projecting different media types into a shared latent space, the model learns manipulation-invariant representations and supports zero-shot detection on unseen modalities. Experiments demonstrate state-of-the-art generalization and outperform conventional CNN-based classifiers by up to 15% on cross-domain benchmarks. However, modality fusion increases training complexity, and fine-grained localization of tampering is limited due to the holistic embedding strategy.

The work [16] leverages vision-language models (VLMs), such as CLIP and BLIP 2, to detect AI-generated images through multimodal contrastive supervision. Instead of training discriminators from scratch, the proposed method prompts pretrained VLMs with textual descriptions emphasizing authenticity cues, which are then aligned with image embeddings. The authors fine-tune the models using a curated dataset

consisting of images generated by Midjourney, DALL-E, Stable Diffusion, and real photographs from LAION and COCO. Results show improved detection accuracy and robustness to domain shifts compared to CNN baselines like EfficientNet and XceptionNet. The system exhibits interpretability through attention heatmaps derived from cross-modal alignment. However, performance drops when encountering entirely new generator architectures, and text prompt engineering remains crucial for high accuracy, limiting automation and scalability.

The paper [17] investigates audio deepfake detection using a self-supervised

contrastive learning framework that minimizes dependence on labeled data. The authors leverage wav2vec 2.0 and BYOL-style pretraining to learn generalizable latent representations from large-scale unlabeled speech corpora. During fine tuning, a projection head and lightweight classifier distinguish between bona fide and spoofed speech generated by TTS and voice conversion systems. Experiments conducted on ASVspoof 2019 LA and VSDC demonstrate that contrastive pretraining improves robustness under cross-dataset evaluation while requiring substantially fewer annotations. The model outperforms spectrogram-based CNNs in low-resource settings and remains resilient to channel noise, bitrate compression, and replay artifacts. However, reliance on large unlabeled corpora increases training time, and the framework offers limited interpretability of decision boundaries.

The work [18] introduces a phase-based deepfake detection approach that captures artifacts overlooked by magnitude-only spectrogram analysis. The authors compute group delay, instantaneous frequency, and modified group delay features from raw speech signals and feed them into a multi-branch convolutional network. Evaluations on ASVspoof 2019, In-the-Wild, and an internal dataset containing unseen vocoders show improved detection of advanced TTS models such as WaveGlow and HiFi GAN. The system demonstrates strong generalization to compression artifacts and background noise, outperforming magnitude-only baselines. While the inclusion of phase information enhances robustness, the increased feature extraction complexity and susceptibility to phase-wrapping errors may affect scalability and real-time deployment.

The paper [19] introduces Skyra, a specialized multimodal large language model (MLLM) developed to provide interpretable AI-generated video detection through grounded artifact reasoning. To address the transparency limitations inherent in traditional binary classifiers, the authors present ViF-CoT-4K, the first large-scale dataset of AI video artifacts featuring fine-grained human annotations and a three layer hierarchical taxonomy covering both perceptual forgeries and physical violations. The framework utilizes a robust two-stage training strategy: a cold-start supervised fine-tuning (SFT) phase to initialize spatio-temporal artifact perception, followed by reinforcement learning (RL) via Group Relative Policy Optimization (GRPO)

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to refine self-driven reasoning capabilities. Experimental results on the ViF-Bench and GenVideo benchmarks demonstrate that Skyra substantially outperforms current state-of-the-art binary detectors and general-purpose MLLMs, achieving an absolute accuracy improvement of 26.73% over established baselines. Despite its superior performance, the study acknowledges that the model's reasoning remains domain bound to specific generator distributions and that its natural-language rationales are still susceptible to the risks of overconfidence and partial hallucination

The paper [20] focuses on zero-shot detection of AI-generated videos by leveraging latent traces left by diffusion-based generation pipelines. The authors analyze spatiotemporal inconsistencies and distributional deviations in denoising trajectories, extracting forensic cues using 3D CNNs and frequency filters. Unlike conventional supervised models, the framework does not require paired training samples for each generator. Instead, it models the intrinsic energy patterns observed in real video temporal evolution and flags deviations. Cross-generator evaluation on Veo2, Kling, Runway, Pika, and Sora shows strong generalization and reduced false-positive rates, even under compression and upscaling. Limitations include weaker performance on low-motion scenes and dependence on higher video resolution for reliable trace extraction.

The work [21] proposes a unified multimodal detection framework capable of identifying deepfakes across images, audio, and video without relying on modality specific architectures. The system employs a shared transformer backbone coupled with modality adapters that project features from spectrograms, video frames, and static images into a common latent embedding space. Training is conducted using a combination of contrastive and cross-entropy losses on datasets such as FaceForensics++, DFDC, ASVspoof 2019, and Stable Diffusion outputs. The model supports cross-modal generalization and zero-shot detection on previously unseen manipulation types. Experimental results show improved robustness over standalone models and strong performance under compression and noise. However, increased training cost and reduced fine-grained localization remain limitations due to modality abstraction.

The paper [22] presents a hybrid detection approach that focuses on the characteristic frequency distributions and patch-level artifacts left by diffusion-generated media.

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The authors design a dual-branch network combining discrete cosine transform (DCT) statistics with patch-based transformers trained on LAION, Midjourney, Stable Diffusion, and DALL·E-generated samples. Their method generalizes well to diffusion based video and image synthesis by emphasizing high-frequency inconsistencies and local texture irregularities. Tests conducted on GenImage, DiffusionForensics, and proprietary datasets show superior cross-model accuracy compared to CLIP-based and CNN baselines. Although the system can detect images and individual video frames, full spatiotemporal modeling is limited, and adversarially smoothed outputs reduce detection confidence.

The study [23] introduces an adversarial fine-tuning strategy to improve resistance against evasion attacks on deepfake detectors. The authors generate adversarial perturbations using PGD and FGSM methods on FaceForensics++, Celeb-DF, and DFDC samples, then fine-tune a Vision Transformer to tolerate pixel-space and latent-space manipulations. The resulting model achieves lower attack success rates while maintaining high clean-data performance. Cross-dataset evaluations also demonstrate resilience to unknown generators such as Make-A-Video and Dreamix. Nonetheless, increased inference time and partial performance degradation under heavy compression are acknowledged challenges. The technique is primarily designed for image and frame-based detection rather than long-form video.

The paper [24] proposes a multi-scale temporal modeling framework that captures both short- and long-term inconsistencies in manipulated videos. The model employs a hierarchical 3D convolutional backbone fused with temporal transformers to detect subtle irregularities in lip-sync, blinking, and motion continuity. Training is performed on FaceForensics++, DeeperForensics, and a new synthetic set derived from text-to-video generative models. The authors report state-of-the-art performance on compressed and low-resolution videos, outperforming SlowFast, Xception-3D, and MVIT. The framework also supports clip-level and video-level predictions. However, real-time deployment remains challenging due to high GPU memory usage, and low motion scenes may reduce detection accuracy.

The work [25] introduces an explainable audio-visual detection system that employs cross-modal attention to identify mismatches between speech and facial

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movements in synthetic media. The model extracts lip-region frames and mel spectrograms, aligns them via attention maps, and learns spatiotemporal inconsistencies. Evaluations on FakeAVCeleb, AVSpeech-Deepfake, and DFDC show improved accuracy over unimodal detectors, especially for partially manipulated content. The system generates interpretable heatmaps visualizing regions of misalignment between audio and visual cues. While the method enhances robustness to modality-specific attacks, it struggles with silent or voice-over scenarios and requires high-quality facial region extraction during preprocessing.

2.1 Research Gap

Although significant progress has been made in the field of deepfake detection, several challenges remain unresolved in existing systems. Most research efforts focus primarily on improving binary classification accuracy using deep learning architectures such as Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). However, many of these systems still face practical limitations when applied to real world, in-the-wild media verification.

One major limitation observed in existing studies is the lack of generalization across unseen generative models. Many proposed approaches are trained on specific, limited public datasets and struggle to detect sophisticated synthetic media produced by newer generative adversarial networks (GANs) or diffusion-based techniques. As a result, the models fail to reflect real-world diversity and struggle with out-of distribution data.

Another challenge is the black-box nature of advanced deep learning models. High performing architectures often do not provide explanations for their predictions, which significantly reduces user trust and usability in critical applications such as journalism, legal forensics, and digital identity authentication.

Additionally, many systems focus on single-modality detection, analyzing either strictly visual artifacts or audio anomalies. This singular focus prevents systems from catching complex, multi-modal manipulations, such as lip-sync mismatches or temporal inconsistencies between video and audio streams.

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Existing research also highlights the heavy computational cost associated with state-of-the-art detection systems. Advanced models require powerful hardware and rely exclusively on cloud processing. This not only limits their suitability for real time applications but also raises severe privacy concerns, as users are forced to upload potentially sensitive data to external servers.

Furthermore, while current systems are designed to detect media manipulation, they lack the capability to verify the authenticity of the information itself. Detecting a deepfake does not automatically debunk the false claims associated with it, leaving a gap in comprehensive misinformation combat strategies.

The proposed TRUTH-SEEKER system attempts to address several of these research gaps. Specifically, it focuses on developing a hybrid, multi-modal framework that integrates visual and audio analysis, explainable AI (XAI) for transparency, edge-cloud computing for privacy preservation, and Retrieval-Augmented Generation (RAG) for fact verification within a unified pipeline.

2.2 Key Insights from Literature

The reviewed literature collectively highlights significant progress in the field of deepfake detection, driven by advances in deep learning, computer vision, and multi modal analysis. These studies emphasize the transition from simple artifact detection to complex spatial-temporal modeling and semantic reasoning to combat highly realistic synthetic media. The overarching goal across these works is to create robust, interpretable, and efficient systems to maintain digital trust.

Modeling Approaches

Recent research demonstrates a clear transition from conventional image-based forensic tools to sophisticated deep learning architectures. CNN-based models are widely used for spatial feature extraction to detect textures and inconsistencies in facial regions. Meanwhile, temporal models and Vision Transformers (ViTs) are increasingly adopted to capture long-range dependencies and sequential inconsistencies across video frames, improving adaptability to new deepfake techniques.

Multi-Modal Analysis

A recurring theme in modern detection research is the shift away from single modality systems. Modern research emphasizes combining multiple data sources, such as video, audio, and text. Multi-modal systems improve reliability by detecting cross-modal inconsistencies, such as lip-sync mismatches and audio anomalies, which unimodal systems often miss during sophisticated manipulations.

Explainable AI (XAI)

Several studies emphasize the importance of Explainable Artificial Intelligence (XAI) in deepfake detection. Because deep learning models are often considered black boxes, integrating techniques such as Grad-CAM, SHAP, and attention visualizations helps highlight the specific regions responsible for classification. This transparency improves user trust and provides actionable insights for media verification.

Hybrid Architectures and Efficiency

The trade-off between accuracy and computational efficiency is another important research focus. High-accuracy models often require substantial computational resources. To address this, recent literature proposes hybrid architectures combining edge and cloud computing. These setups allow lightweight models to perform quick, privacy-preserving detection locally, while complex models operate in the cloud for detailed analysis.

Fact Verification Integration

Recent advancements also explore the integration of fact-verification mechanisms. Detecting manipulated media alone is insufficient to combat misinformation; verifying associated claims is equally important. Retrieval-Augmented Generation (RAG) and knowledge-based systems have been identified as effective solutions to retrieve trusted information and generate evidence-based explanations.

Datasets and Generalization Challenges

The reviewed studies highlight that detection effectiveness heavily depends on the quality and diversity of training datasets. Data augmentation and transfer learning are frequently utilized to improve model robustness, emphasizing that continuous data updates are essential for maintaining high performance against unseen generative techniques. Models trained exclusively on a single dataset often experience significant

performance degradation when exposed to deepfakes generated by novel algorithms or different demographic distributions.

2.3 Formulation of the Problem

Despite significant progress in deepfake detection technologies, identifying manipulated media and verifying digital authenticity in real-world scenarios remains a challenging problem. The rapid evolution of generative AI makes it increasingly difficult to distinguish between real and fake content using traditional methods. Furthermore, the lack of transparency, heavy computational requirements, and the inability to contextually verify facts add additional complexity to automated detection systems. These challenges highlight the need for intelligent systems capable of providing comprehensive, private, and trustworthy media analysis.

2.3.1 Problem Statement

The increasing availability of AI-generated media and the rapid growth of deepfake technology have created significant challenges in maintaining the authenticity and reliability of digital media. Although several research

efforts have proposed deepfake detection systems, many existing tools suffer from several limitations. These limitations include:

- Poor generalization to unseen deepfake techniques due to reliance on biased or limited training datasets.
- A black-box nature that fails to provide clear explanations for predictions, reducing user trust in critical scenarios.
- A single-modality focus that misses complex, cross-modal manipulations like audio-visual desynchronization.
- High computational demands and privacy risks, as most systems require cloud processing and force users to upload sensitive media.
- An inability to perform fact verification, meaning systems can flag manipulated media but cannot verify or debunk the associated contextual claims.

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These limitations reduce the practicality of current solutions for real-world, real time media verification. Therefore, there is a clear need to develop a system that can accurately detect deepfakes while providing explainable results, handling multiple data modalities, and ensuring user data privacy.

2.3.2 Aim of the Project

The aim of this project is to design and develop a hybrid deepfake detection system that integrates multi-modal analysis, explainable AI, and privacy-preserving mechanisms to ensure accurate and reliable detection of manipulated content.

The system focuses on building an efficient verification pipeline that captures media input, extracts spatial and temporal features across visual and audio modalities, and classifies them using advanced deep learning models. In addition, the system incorporates Retrieval-Augmented Generation (RAG) to process contextual claims and provide fact-checked, evidence-based explanations alongside visual heatmaps. By integrating these components via an edge-cloud architecture, the project aims to provide a secure, transparent, and robust tool against digital misinformation.

2.3.3 Objectives

In order to achieve the above aim and address the identified research gaps, the project is guided by the following specific objectives. Each objective reflects a clear, implementable task that contributes to the overall solution.

- To develop a multi-modal detection module capable of simultaneously analyzing image, video, and audio streams to identify cross-modal inconsistencies indicative of manipulated media.
- To integrate Explainable AI (XAI) techniques, specifically utilizing saliency maps and attention visualization, to improve system transparency by explicitly highlighting manipulated regions within the media.
- To implement an automated fact-checking pipeline leveraging Retrieval-Augmented Generation (RAG) architecture, which will systematically extract textual claims,

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transcripts, and contextual metadata associated with the analyzed media, dynamically cross-referencing this information against curated, highly reliable external knowledge bases and real-time repositories to accurately validate claims and expose contextual incongruities.

- To design a privacy-aware hybrid architecture that combines edge and cloud computing paradigms, enabling the local processing of sensitive data to ensure user privacy while maintaining computational efficiency.
- To train the detection models utilizing diverse, cross-domain datasets and advanced data augmentation techniques, thereby improving the system's generalization capabilities against novel or unseen deepfake generation methods.
- To evaluate the proposed system comprehensively using quantitative metrics such as detection accuracy, false positive rates, computational latency, and explanation precision to ensure real-world viability.
- To demonstrate and document the applicability of the integrated system in practical domains such as social media moderation, digital forensics, and news verification, establishing a reliable foundation for combating misinformation.

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Chapter 3

Proposed Methodology

3.1 Overall System Architecture

The proposed system, TRUTH-SEEKER, is designed as a hybrid multi-modal framework that enables deepfake detection and contextual fact verification. The architecture integrates computer vision, multimodal large language models, and edge-cloud computing techniques to facilitate secure, accurate, and explainable media authentication for end users.

The system operates through two primary processing stages: Edge-Based Triage and Cloud-Based Deep Analysis. These stages work together to support efficient, privacy-preserving, and highly accurate multi-modal verification. The overall architecture combines lightweight local inference, advanced visual encoding, natural language reasoning, and an interactive explainable user interface to produce meaningful outputs for complex media.

In the Edge-Based Triage stage, the system receives user-uploaded video through a mobile application built with Kotlin and Jetpack Compose. The media is immediately processed by an On-Device Triage Model, which acts as the first line of defense. This lightweight edge AI scans the visual data to quickly flag glaring manipulation artifacts. If the video passes this initial check, it can be marked as authentic locally, preserving user privacy and saving network bandwidth. However, if the media exhibits suspicious artifacts, it is securely forwarded to the cloud for a more rigorous inspection.

In the Cloud-Based Deep Analysis stage, the forwarded video is received by a

GPU-accelerated FastAPI backend via secure HTTPS protocols. The system extracts sequential frames from the video, which are then processed by a Qwen2.5-VL Visual Encoder. This advanced encoder analyzes both the two-dimensional spatial layout of the images and the three-dimensional temporal flow across the video sequence. The extracted representations are fed into a Visual Chain-of-Thought (CoT) Multimodal Large Language Model. Instead of producing a black-box score, this model utilizes a Joint Classifier to mathematically balance classification loss with language modeling loss, generating a binary verdict alongside a natural language explanation and precise spatial coordinates (bounding boxes) of the manipulation.

The architecture also includes supporting modules such as an automated fact verification pipeline and an Explainable Results UI. Once the cloud analysis is complete, the backend compiles the binary verdict, textual reasoning, and spatial data into a detailed JSON report. This report is transmitted back to the mobile

application, where the UI renders a visual evidence overlay directly onto the video, allowing users to visually interpret the specific anomalies and textural inconsistencies identified by the system.

Figure 3.1: Architecture of the Hybrid Deepfake Detection. Figure 3.1 illustrates the TRUTH-SEEKER architecture, dividing the workload between lightweight mobile edge triage and advanced cloud analysis. Suspicious media is securely escalated to the DAVID-XR1 cloud pipeline, which utilizes a Vision Language model to generate an explainable verdict. These findings are then returned to the mobile application as a report to display an interactive visual evidence overlay.

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3.2 Technology Stack

The system is built using a combination of modern programming languages, frameworks, and tools to support multi-modal processing and real-time performance. These technologies are carefully selected to ensure efficient handling of diverse data types such as images, videos, audio signals, and textual inputs, while maintaining scalability and robustness.

The system leverages powerful frameworks such as PyTorch and TensorFlow to develop deep learning models used in deepfake detection. PyTorch provides flexibility and ease of experimentation for model training, while TensorFlow and TensorFlow Lite are used for optimizing and deploying models, particularly on edge devices where computational resources are limited. OpenCV is utilized for efficient image and video processing tasks such as frame extraction, preprocessing, and transformation, which are essential for visual analysis.

For audio processing, the system uses Librosa to extract important features such as frequency components and spectrograms, enabling effective detection of anomalies in audio signals. Additionally, the Whisper model is integrated for speech-to-text conversion, allowing the system to process and analyze spoken content. These components together enable comprehensive multi-modal analysis by combining visual and audio information.

To facilitate seamless user interaction on edge devices, the mobile application frontend is developed using Android Studio. It leverages Kotlin as the primary programming language and Jetpack Compose for building a modern, responsive, and native user interface. This frontend integrates directly with the on-device inference models and communicates securely with the cloud infrastructure.

FastAPI is employed to develop high-performance backend APIs that facilitate communication between these mobile edge devices and the cloud servers. It ensures fast data transfer and supports real-time interaction. Cloud platforms such as Firebase or other cloud services are used to provide scalable storage, authentication, and computational resources required for advanced processing tasks.

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To improve system efficiency, model optimization techniques such as quantization and pruning are applied. These techniques reduce the size of machine learning models and enhance inference speed, making them suitable for real-time deployment on resource constrained devices. This optimization is particularly important for the edge module, where low latency and fast response times are critical.

The system follows a modular and API-driven architecture, enabling seamless integration between different components. Secure communication protocols are implemented to ensure data privacy and integrity during data transmission between edge and cloud modules.

Overall, the chosen technology stack provides a balanced combination of performance, scalability, and efficiency, enabling the system to handle complex multi-modal data while delivering accurate and real-time deepfake detection and fact verification.

Programming Languages

- Python – Used as the primary language for backend architecture, facilitating the training, validation, and deployment of complex deep learning models due to its extensive ecosystem and robust scientific computing libraries.
- Kotlin – Selected as the modern, statically typed programming language for native Android mobile application development, offering enhanced performance, null safety, and seamless interoperability with existing Java ecosystems.

Libraries and Frameworks

- PyTorch – Employed as the core framework for designing, training, and fine tuning the deep learning architectures (such as ViTs and CNNs), chosen for its dynamic computation graph and ease of experimentation during the research phase.
- TensorFlow / TensorFlow Lite – Utilized specifically for the quantization, compression, and deployment of trained models onto edge devices, ensuring low-latency inference and a minimal memory footprint on mobile hardware.

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- OpenCV – Integrated for high-performance computer vision tasks, including real-time video frame extraction, spatial normalization, color space conversion, and essential image preprocessing pipelines required prior to model inference.
- Librosa – Leveraged for advanced audio signal processing, enabling the extraction of critical acoustic features such as Mel-frequency cepstral coefficients (MFCCs), spectrogram generation, and high-frequency anomaly detection.
- Whisper – Integrated as the state-of-the-art automatic speech recognition (ASR) system to accurately transcribe spoken audio into text across multiple languages, facilitating downstream natural language analysis and fact verification.
- FastAPI – Adopted for constructing high-performance, asynchronous backend RESTful APIs, ensuring rapid, secure, and efficient data transmission between the mobile edge clients and the cloud-based processing servers.
- Jetpack Compose – Utilized as the modern declarative native UI toolkit for Android, enabling the rapid development of a highly responsive, intuitive, and interactive user interface that can smoothly render explainable AI heatmaps and dynamic verification results.

Platforms and Tools

- Android Studio – Serves as the primary Integrated Development Environment (IDE), providing essential debugging tools, emulators, and profiling capabilities required for robust mobile application engineering.
- Firebase / Cloud Services – Deployed for secure user authentication, scalable real-time database management, and robust cloud storage solutions to handle the secure archiving of uploaded media and detailed JSON verification reports.
- GitHub – Utilized for distributed version control, enabling structured team collaboration, continuous integration pipelines, and systematic tracking of the project's multi-modal codebase evolution.

The selected technology stack ensures high performance, scalability, and ease of deployment across different platforms.

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3.3 Design Details

The design of the proposed TRUTH-SEEKER hybrid deepfake detection system integrates multiple processing modules that collectively enable real-time, privacy preserving media analysis. The system is designed as a modular edge-cloud architecture where each component performs a specific task such as fast frequency triage, deep temporal analysis, explainable AI generation, and semantic verification. By combining lightweight mobile frameworks with heavy GPU-accelerated vision-language models and retrieval-augmented generation, the system is capable of detecting and explaining synthetic media with high accuracy.

The system operates through two main processing zones. The first zone is the mobile edge application, which performs rapid, on-device processing of video and audio streams to catch obvious synthetic artifacts and maintain user data privacy. The second zone is the secure cloud backend, which performs deep, multi-modal analysis and generates explainable reports for media that fails the initial edge triage.

To effectively process these diverse media formats, the TRUTH-SEEKER framework delegates specific tasks to highly specialized neural network architectures within these zones. The detailed working of this overall hybrid architecture is illustrated in the figure. The following sections detail the individual components and operational considerations that make up this ecosystem.

Figure 3.2: Hybrid Deepfake Detection Processing Pipeline. 28

3.3.1 Video Pipeline

The video processing pipeline is designed to analyze spatial manipulation and temporal inconsistencies, operating across both the user's device and the cloud backend.

Edge (Triage): Frequency Stream Processing

Running a frequency model on the Kotlin/Jetpack Compose mobile module allows for fast, lightweight spatial screening to catch obvious artifacts directly on the device before any data is sent over the network.

Figure 3.3: ConvNeXt Nano architecture. The visual frequency stream at the edge is processed using a lightweight ConvNeXt Nano architecture. It acts as a modernized ResNet that incorporates Vision Transformer (ViT) design principles while remaining a purely Convolutional Neural Network with approximately 15 million parameters. This module acts as the system's "Texture Inspector," specifically looking for invisible digital forensics such as checkerboard patterns, noise residuals, and GAN fingerprints left by the generation process.

Instead of traditional temporal processing, 1D time-domain signals are converted into 2D time-frequency representations, such as Spectrograms. The architecture utilizes a 4-stage hierarchical design where spatial resolution decreases while the channel dimension increases. It features a "patchify" stem using 4×4 convolutions, large 7×7 depthwise kernels for robust global context, and modern GELU/LayerNorm components. This allows ConvNeXt to seamlessly process variable-length frequency

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streams by treating them as single-channel images, detecting patterns across time and frequency simultaneously before passing them to a global average pooling and fully connected classifier to make a local confidence decision.

Cloud (Deep Inspection): Temporal and Cross-Modal Analysis

Pushing the heavy lifting to the GPU-accelerated backend facilitates deep, transparent, and explainable multi-modal evidence generation for files that fail edge triage. The cloud utilizes an ensemble approach, running a deeper pass of ConvNeXt Nano alongside specialized temporal and explainable models.

Figure 3.4: TimeSformer architecture. To detect temporal anomalies and inter-frame inconsistencies, the cloud employs the TimeSformer (Time-Space Transformer) architecture. Unlike static image generators, deepfake video synthesis often struggles to maintain strict temporal coherence, resulting in subtle flickering, unnatural motion dynamics, or temporal blending errors across consecutive frames. The TimeSformer is explicitly designed to capture these sequential artifacts without the heavy computational burden of traditional 3D Convolutional Neural Networks.

The pipeline begins by extracting a continuous sequence of video frames. Instead of processing entire video volumes simultaneously, the TimeSformer divides the video sequence into non-overlapping spatio-temporal patches. The core of this network relies on a "Divided Space-Time Attention" mechanism.

For each extracted patch, the model first applies Temporal Self-Attention, comparing a specific spatial patch exclusively with its corresponding patches in past and future frames. This

directly tracks local pixel regions over time, isolating unnatural jitter or temporal degradation. Immediately following this, the model applies Spatial Self-Attention to analyze the relationships between different patches within the exact same frame. By decoupling the spatial and temporal dimensions, the TimeSformer efficiently learns robust representations of natural video flow.

Figure 3.5: DAVID-XR1 Vision-Language Multimodal Model. Finally, the outputs from TimeSformer and ConvNeXt are fed into the DAVID XR1 framework, built upon the Qwen2.5-VL-7B Vision-Language Multimodal Model (VLMM). This provides the deep, explainable multi-modal evidence generation needed for complex deepfakes. It utilizes an advanced "MROPE" mechanism to understand 3D space (adding the element of time for video frames) and undergoes CoT-Aware Supervised Fine-Tuning to output a structured sequence: a step for reasoning, an generation step, and a final verdict. The network is optimized using a joint loss function:

$$L = \alpha LLM + \beta LCLS$$

3.3.2 Audio Pipeline

The audio processing pipeline is dedicated to identifying synthetic voice generation and fully AI-generated music, utilizing a similar split between rapid local assessment and deep cloud verification. Specifically, the module evaluates features such as mel spectrograms and frequency artifacts to distinguish authentic human speech from algorithmically synthesized sounds.

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Edge (Triage): Fast Spectrum Triage

For the mobile module, keeping the on-device footprint small while still effectively catching initial synthetic frequency signals is paramount. The edge triage utilizes the Spectra Gamma or Beta models. These lightweight variants of the Spectra architecture process abbreviated Mel-spectrograms locally. By performing a rapid pass over the audio frequencies, the edge application can confidently clear benign voice notes or flag suspicious synthetic audio for escalation without draining the mobile device's battery or compromising user privacy.

Cloud (Deep Inspection): Raw Waveform Analysis

When the edge models flag a suspicious voice or audio clip, the file is escalated to the cloud. Ramping up to the heavy-duty Wav2Vec 2.0 Large on the GPU backend ensures a robust, highly accurate analysis directly from the source audio.

Figure 3.6: Wav2Vec 2.0 architecture. To combat the rise of highly realistic voice cloning, Wav2Vec 2.0 bypasses traditional, human-crafted Mel-spectrograms entirely. Instead, it ingests the raw 1-dimensional audio signal at 16,000 Hz, passing it through a 7-layer Convolutional Neural Network (CNN) feature extractor. This

continuous compression translates microscopic audio vibrations directly into dense mathematical feature vectors, preserving subtle phase anomalies and structural glitches that are often smoothed over or discarded by spectrogram conversions.

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A massive 24-layer Transformer context network then applies multi-head self attention to these learned features. This allows the model to deeply evaluate temporal coherence, capturing long-range acoustic dependencies and unnatural synthesis artifacts across the entire speech sequence. Finally, a specialized binary classification head replaces the standard quantization modules, aggressively scoring the structural integrity of the temporal flow to detect even the most sophisticated digital manipulation.

3.3.3 Semantic Verification Pipeline

The semantic verification pipeline operates to analyze text-based claims, transcriptions from the audio pipeline, and broader narrative context. Because this relies heavily on large-scale databases and memory-intensive models, this pipeline is executed exclusively in the cloud.

Cloud (Fact-Checking): RAG Architecture

Figure 3.7: Retrieval-Augmented Generation (RAG) pipeline. To verify claims and text-based rumors, the cloud system utilizes a Retrieval Augmented Generation (RAG) architecture. Standard LLMs rely purely on internal parametric memory—compressing training data into statistical weights. If a user asks about a highly specific or recent fact where these weights are weak, the model will often hallucinate a plausible but fabricated response. RAG shifts the paradigm to treat the LLM strictly as a reasoning engine rather than a closed knowledge base.

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The RAG implementation operates in three distinct phases within the cloud:

1. Active Retrieval: Before the LLM is activated, the backend acts as an autonomous agent, taking the user's query and fetching external, verified data from vector databases using semantic similarity searches.
2. Context Window Augmentation: The system carefully constructs a prompt that fits within the model's memory limits by stitching together strict System Instructions, the Retrieved Evidence, and the User's Query.
3. Grounded Generation: Because the system instructions explicitly force the LLM to use only the provided evidence, the model's internal token probabilities shift dramatically. The probability of hallucinating drops to near zero, resulting in an answer securely grounded in verifiable facts.

3.3.4 System Efficiency

To achieve real-time, multi-modal performance in computationally constrained environments, the TRUTH-SEEKER framework employs several advanced architectural and algorithmic optimization strategies. These ensure the system remains highly responsive under load:

- Edge-Cloud Load Balancing: By utilizing lightweight edge models to perform a rapid initial triage, the system acts as a dynamic filter. It offloads only highly suspicious or ambiguous media to the cloud, drastically reducing unnecessary network payload, minimizing user bandwidth consumption, and lowering overall cloud compute expenditures.

- **Model Compression for Edge Deployment:** To ensure smooth operation on diverse mobile hardware, the system leverages techniques such as Post-Training Quantization (PTQ)—converting standard FP32 weights to INT8—and targeted weight pruning. These strategies significantly reduce the memory footprint and accelerate on-device inference speed without suffering a substantial degradation in classification accuracy.

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- **Cloud Hardware Acceleration:** The cloud backend maximizes parallel processing and GPU utilization by deploying models within highly optimized execution environments like NVIDIA TensorRT. Coupled with dynamic request batching, the system efficiently pools concurrent analysis requests, maximizing throughput and maintaining stability during high-traffic spikes.

- **Asynchronous Processing and Queuing:** Deep temporal and multi-modal analysis are inherently time-consuming. To address this, the backend utilizes asynchronous API endpoints integrated with robust message brokers. This architecture handles heavy video processing in the background without blocking the main execution threads, allowing the mobile client to remain responsive while polling for the final report, thereby minimizing perceived latency for the end user.

Ultimately, this hybrid architectural design seamlessly balances the inherent trade offs between processing latency, power consumption, and analytical performance. By intelligently routing computational tasks to their optimal environments, the system guarantees both rapid on-device triage and high-fidelity, forensic-grade deep analysis at scale.

3.3.5 Security and Privacy Considerations

Given the sensitive nature of user media, the architecture adheres to a strict privacy by-design philosophy, implementing robust security at both the edge and cloud levels:

- **Data Minimization:** Benign media is processed locally and never leaves the user's device; only highly suspicious content is transmitted for deep analysis.
- **Ephemeral Cloud Processing:** Uploads and intermediate artifacts are processed strictly in volatile memory and permanently purged immediately after the final report is generated.
- **End-to-End Encryption:** All data transmission is secured using TLS 1.3, while any necessary temporary caching utilizes AES-256 encryption at rest.

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- **Adversarial Robustness:** Internal models are trained with adversarial augmentations to resist evasion attacks, preventing malicious actors from bypassing detection.
- **Explicit User Consent Mechanisms:** The escalation of flagged media to the cloud tier strictly requires unambiguous, opt-in consent from the user, guaranteeing complete transparency regarding data transmission and processing.
- **Edge Model Integrity:** The lightweight detection models deployed locally are secured via cryptographic signing to prevent reverse engineering, unauthorized tampering, or model extraction attacks on the client side.

These comprehensive measures ensure the system strictly adheres to global data protection regulations while maintaining a highly secure operational environment.

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Chapter 4

Implementation and Experimental Investigation

4.1 Dataset

The performance of a deepfake detection system depends heavily on the quality and diversity of the datasets used for training and evaluation. In this project, multiple datasets containing real and AI-generated media are utilized to train and validate the detection models.

The dataset includes samples from different modalities such as images, videos, and audio clips. Each sample is labeled as Real or Fake, enabling supervised learning for classification tasks.

To ensure robustness and generalization, the dataset contains variations in lighting conditions, compression levels, generation techniques such as GANs and diffusion models, as well as noise and distortions. This diversity helps the model learn generalized features rather than overfitting to specific patterns.

In addition to benchmark datasets such as FaceForensics++ and DFDC, custom collected samples are also included to simulate real-world scenarios. These samples are obtained from publicly available sources such as social media platforms and news datasets, enabling the system to handle real-world deepfake content effectively.

The dataset is divided into three subsets: Training Set (70%) for learning model 37

parameters, Validation Set (15%) for tuning hyperparameters, and Test Set (15%) for evaluating final model performance. This structured split ensures reliable and unbiased evaluation.

Data augmentation techniques are applied to increase dataset diversity and prevent overfitting. For images and videos, techniques such as rotation, flipping, cropping, brightness adjustment, and noise addition are used. For audio data, pitch shifting, time stretching, and background noise injection are applied. These transformations improve model robustness under varying conditions.

To handle class imbalance between real and fake samples, balancing strategies such as oversampling and weighted loss functions are employed. This ensures that the model does not become biased toward a particular class.

The dataset includes both low-quality and high-quality deepfake samples. Low quality deepfakes often contain visible artifacts, while high-quality deepfakes are more realistic and difficult to detect. Including both types ensures that the model is trained for a wide range of manipulation complexities.

Temporal consistency is also considered for video datasets. Deepfake videos often exhibit frame-level inconsistencies such as flickering and unnatural transitions. By analyzing sequential frames, the system learns to detect these temporal irregularities.

Audio datasets include both natural speech and synthetic speech generated using various text-to-speech and voice cloning techniques. This enables the model to distinguish between real and generated audio signals effectively.

Metadata such as compression format, resolution, and encoding techniques is also considered during preprocessing. These additional attributes provide useful cues for identifying manipulated content.

4.1.1 Training Dataset Preparation

Before the model training phase commences, all multi-modal media inputs must undergo a rigorous sequence of preprocessing and feature extraction. This stage is critical

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for standardizing the heterogeneous raw data, minimizing irrelevant environmental noise, and highlighting the subtle digital artifacts indicative of manipulation.

- **Image and Video Data:** For video inputs, temporal sequences are first parsed, and individual frames are extracted at a standardized frame rate to ensure temporal consistency across the dataset. These frames, along with static image inputs, are subsequently resized to a uniform resolution (e.g., 256×256 pixels) and normalized to scale pixel intensities appropriately. To ensure the detection model focuses exclusively on the most frequently manipulated regions, advanced face detection algorithms are deployed. These algorithms locate, align, and crop the facial regions, effectively discarding irrelevant background information. This targeted cropping brings specific attention to spatial inconsistencies, unnatural blending boundaries, and spatial warping artifacts present in synthetically generated faces.

- **Audio Data:** Raw audio tracks are first resampled to a consistent sampling rate to ensure uniformity across the training corpus. The one-dimensional temporal audio signals are then mathematically transformed into two-dimensional visual representations, specifically Mel-spectrograms, to capture both temporal and spectral characteristics. Prior to further analysis, the audio data undergoes rigorous ambient noise reduction and amplitude normalization. These transformations neutralize volume discrepancies and background interference, yielding clean, standardized acoustic sequences that are highly optimized for ingestion by deep learning architectures.

Following the foundational preprocessing phase, specialized feature extraction pipelines are utilized to derive high-dimensional representations from the standardized data. For the visual modality, robust convolutional architectures, such as pretrained ResNet models, are employed to extract deep spatial hierarchies. These networks are highly adept at capturing pixel-level anomalies, illumination mismatches, and micro-texture inconsistencies that are inherently left behind by deepfake generation techniques. Conversely, for the auditory modality, Mel-Frequency Cepstral Coefficients (MFCCs) are utilized to extract vital acoustic features. These coefficients meticulously capture

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phonetic characteristics, unnatural frequency variations, and synthetic vocal tract anomalies. Ultimately, these enriched visual and auditory feature sets provide the highly discriminative inputs required for the system's cross-modal detection and alignment architecture.

4.2 Hyperparameters

The deep learning models used in this project require several hyperparameters to control training and performance. The learning rate determines how quickly the model updates its weights, while batch size controls the number of samples processed in each iteration.

The number of epochs defines how many times the model is trained on the dataset. The Adam optimizer is used for efficient convergence, and binary cross-entropy loss is applied for classification tasks. Dropout is used as a regularization technique to prevent overfitting by randomly disabling neurons during training.

Techniques such as early stopping and learning rate scheduling are used to improve training efficiency and prevent overfitting. Cross-validation is also performed to ensure reliable performance.

Model Training and Evaluation

The system uses multiple deep learning models for detecting deepfakes across different modalities, guided by the aforementioned hyperparameters.

CNN and ResNet-based models are used for image detection, temporal models are used for video analysis, and spectrogram-based models are used for audio detection. These models are trained using labeled datasets containing real and fake samples.

The performance of the model is evaluated using standard metrics such as Accuracy, Precision, Recall, and F1-score:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

High values of these metrics indicate effective deepfake detection performance. Confusion matrices are also analyzed to understand classification errors and identify challenging cases. Multi-modal evaluation is performed to assess the contribution of each modality to overall system performance.

4.3 Hardware and Software Tools

The development of the TRUTH-SEEKER system utilizes a combination of programming languages, machine learning frameworks, and development tools to ensure efficient processing, scalability, and real-time performance. These technologies support various stages of the system, including deepfake detection, multi-modal analysis, fact verification, the user-facing mobile application, and explainable output generation.

Programming Languages

- Python: Used for implementing deep learning models, data preprocessing pipelines, and backend system logic. It offers an extensive ecosystem of libraries that accelerates research and ensures robust code maintainability across the entire detection pipeline.
- Kotlin: Selected as the primary programming language for engineering the native Android mobile application due to its official support and seamless interoperability. Kotlin's modern, expressive syntax and strict built-in null-safety features significantly mitigate the risk of unhandled runtime exceptions, thereby establishing a highly stable and resilient application architecture. Furthermore,

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its native support for advanced asynchronous programming via coroutines enables the efficient, non-blocking execution of resource-intensive tasks such as handling large media uploads and communicating with backend inference APIs ultimately delivering a fluid, responsive, and user-centric experience.

Frameworks and Libraries

- Jetpack Compose: Employed for building the modern, declarative user interface of the Android application. This approach significantly reduces boilerplate code while ensuring a fluid and highly responsive user experience.
- PyTorch: Used for building and training deep learning models for image, video, and multi-modal analysis. Its dynamic computation graph allows for rapid prototyping and efficient fine-tuning of complex neural networks.
- TensorFlow / TensorFlow Lite: Used for model optimization and deployment, particularly for edge-level AI operations. This ensures that models run efficiently on mobile devices with minimal latency and reduced memory footprints.
- OpenCV: Provides computer vision functionalities such as video capture, frame extraction, and spatial preprocessing. Its highly optimized C++ backend ensures that media files and real-time video streams are handled with maximum efficiency.
- Librosa: Used for advanced audio signal processing and critical feature extraction. Specifically, it enables the generation of Mel-spectrograms required for identifying synthetic vocal anomalies and frequency mismatches.
- FastAPI: Used to develop high-performance backend APIs to facilitate secure communication between the mobile app and cloud models. Its built-in asynchronous capabilities allow the system to handle multiple simultaneous media analysis requests reliably.

Hardware Tools

- GPU-enabled Laptop/Workstation: Used for training deep learning models and performing high-speed cloud inference. These powerful computing machines

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significantly reduce the time required to process and iterate on large-scale multi-modal datasets.

- CPU-based / Mobile Edge Devices: Used for on-device edge-level processing and real-time lightweight triage detection. By processing sensitive media locally, these devices help maintain strict user privacy while conserving network bandwidth.
- Storage Systems: Used for securely storing vast training datasets, metadata, and model checkpoints. These storage architectures are designed with high redundancy to guarantee data integrity and rapid retrieval during training cycles.

Software Tools

- Android Studio: Used as the primary integrated development environment (IDE) for building, testing, and debugging the mobile application. It provides powerful native profiling tools that help optimize the app's performance and battery consumption on diverse mobile hardware.
- Visual Studio Code: Used as the primary development environment for backend architecture and AI model development. Its extensive marketplace of plugins facilitates seamless integration with Python virtual environments, containerization, and version control systems.
- Git and GitHub: Used for strict version control, team collaboration, and overall project management. These platforms enable structured tracking of codebase changes and secure coordination among different development branches.
- Cloud Platforms (Firebase/AWS): Used for scalable remote storage, secure user authentication, and cloud-based heavy processing. They provide the necessary infrastructure to dynamically scale computational

resources based on fluctuating user demand and detection workloads.

Supporting AI Models

- CNN / ResNet Models: Used for detecting deepfake images and extracting hierarchical visual features. Their deep architectural layers excel at identifying

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spatial inconsistencies, micro-textures, and blending artifacts left behind by synthetic generators.

- Temporal Models: Used for analyzing sequential, frame-level inconsistencies within manipulated videos. By evaluating changes over time, they successfully capture unnatural temporal jitter, biological mismatches, and unsynchronized physiological movements.
- Spectrogram-Based Models: Used for detecting sophisticated anomalies in generated or cloned audio signals. They operate on visual representations of sound to spot subtle frequency distortions and acoustic artifacts inherent in AI synthesized voices.
- Whisper: Converts human and synthetic speech into highly accurate text transcripts for further analysis. This automatic transcription establishes the necessary textual foundation required for the subsequent semantic fact-checking pipeline.
- RAG-based System: Performs automated fact verification using a retrieval augmented generation architecture. It cross-references extracted claims and transcripts against trusted external databases to expose contextual inconsistencies and logical fallacies.
- Explainable AI (XAI) Tools: Generate informative heatmaps, saliency maps, and visual bounding boxes overlaid on the media. These visual aids dramatically improve system transparency by explicitly showing end-users which specific regions of the content were mathematically flagged as manipulated.

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Chapter 5

Results and Discussion

This chapter presents the experimental results obtained from the implementation of the proposed TRUTH-SEEKER system. The evaluation focuses on the effectiveness of the user interface, edge-to-cloud routing, multi-modal deepfake detection, and fact verification modules.

The system was evaluated using image, video, audio, and text inputs. The experimental results demonstrate the system's ability to provide low-latency edge scanning, detect deepfake content via complex cloud analysis, and provide reliable, locally processed fact-checking results.

5.1 Mobile Application Interface and User Flow

To ensure accessibility and ease of use, the TRUTH-SEEKER system features a dedicated mobile application that serves as the primary user gateway for both media scanning and textual fact-checking.

From the home interface (Figure 5.1), users can navigate to the Fact Check module, which utilizes a conversational AI interface. This module leverages local AI and RAG (Retrieval-Augmented Generation) approaches to verify text-based claims directly on the device, providing immediate, context-aware responses without requiring heavy cloud computation.

For media verification, users enter the Scan module, where the system architecture's dual-tier processing becomes user-facing. Users are prompted to choose between a Quick Scan and a Deep Scan.

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Figure 5.1: The TRUTH-SEEKER mobile application home screen.

Figure 5.2: The Fact Check AI interface. 46

Figure 5.3: The Scan Type selection screen. The Quick Scan is designed for low latency, utilizing the system's edge-based lightweight models to provide a rapid authenticity prediction. Conversely, the Deep Scan routes the media to the advanced cloud modules for comprehensive multi-modal analysis. Upon selection, users securely upload media from their device. To analyze videos, the app extracts and evaluates a specific sequence (e.g., scanning 30 distinct frames for temporal inconsistencies).

5.2 Multi-Modal Deepfake Analysis

The system employs a highly efficient hybrid edge-cloud architecture for evaluating both audio and video streams. Lightweight triage and initial inference occur at the edge, providing real-time assessments while minimizing bandwidth and preserving user privacy. When the edge models flag high-risk anomalies, or when media requires more intensive computational scrutiny, the content is securely escalated to the cloud based Deep Scan dashboard. Here, advanced cloud modules break down the media into

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specific streams to perform granular, frame-by-frame, and micro-acoustic evaluations to detect complex manipulations. This multi-tiered approach allows the system to cross-reference discrepancies between visual and auditory channels, ensuring that sophisticated cross-modal deepfakes are reliably intercepted. Throughout this process, the user is kept informed via an intuitive interface that dynamically updates as the media progresses from local screening to deep forensic analysis.

5.2.1 Video Analysis

The video analysis pipeline integrates multiple specialized detectors to evaluate both static spatial artifacts and dynamic temporal inconsistencies that edge devices might miss.

Video Frequency Analysis

The system utilizes High-Pass Fourier Transform analysis to detect geometric grid artifacts commonly left behind by Generative Adversarial Networks (GANs) and diffusion models.

When processing an AI-generated video, the cloud system effectively isolates synthetic artifacts. During testing, an AI-generated interior video was flagged with an Average Video Confidence of 97.50% Fake and a Peak Frame Confidence of 100.00% Fake, successfully alerting the user to consistent geometric grid artifacts. Conversely, when an authentic outdoor landscape video was processed, the frequency analysis confirmed the presence of a natural, organic high-frequency distribution, returning a minimal artifact score of 4.05% Fake.

Figure 5.4: Frequency Analysis results. 48

Temporal Deepfake Detector

In addition to static frame analysis, the system rigorously evaluates the physical consistency and temporal coherence of video sequences to catch sophisticated deepfakes that might otherwise pass standard frame-by-frame checks. By utilizing 3D convolutional networks, this temporal module acts as a "Physics Watcher,"

meticulously scanning 16-frame clips for unnatural movements, subtle eye-flickering, and subtle jittering artifacts that are often introduced during the generative process.

When subjected to manipulated media, the temporal module consistently identifies these physics violations and frame-to-frame anomalies with high precision. During our testing phase, an AI-generated video sequence was conclusively flagged, receiving a classification of 99.95% AI Probability (High Risk). Conversely, when analyzing authentic, unmanipulated footage, the temporal module correctly verified the presence of natural temporal coherence and unbroken physical laws, yielding a highly confident Authentic Probability of 99.98%.

Figure 5.5: Temporal analysis results. Explainable AI Integration

To address the inherent "black-box" limitations of traditional deep learning models, the cloud-based video analysis pipeline integrates an advanced Explainable AI (XAI) framework. This integration is powered by the DAVID-XR1 architecture, which is built upon the Qwen2.5-VL-7B Vision-Language Multimodal Model (VLMM). By utilizing a Chain-of-Thought (CoT) reasoning approach and a Joint Classifier, the system

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calculates a binary classification while simultaneously generating natural language explanations and precise spatial coordinates for the detected anomalies. During the experimental evaluation, an AI-generated video (Football Video.mp4) was subjected to the cloud-based Deep Scan module. The system's primary dashboard presented the Forensic Analysis Results, successfully classifying the media as AI Generated with a high-risk confidence score of 98.68%. The interface immediately provided contextual reasoning, explicitly identifying temporal breakdowns and spatial artifacts to justify the deepfake classification.

Figure 5.6: Forensic Analysis Results dashboard. To further demystify these findings, the system provides interactive visualizers. The Temporal Coherence Analysis generated a visual timeline graph explicitly highlighting a sudden breakdown in physics that peaked at Frame 15, indicative of an unnatural "hard glitch." Complementing this, the Spatial Artifact Localization module produced frame-by-frame forensic heatmaps, demonstrating extreme mathematical gradient spikes concentrated precisely around the manipulated subject matter. Additionally, the audio-visual synchronization overlays pinpointed a distinct millisecond discrepancy between the subject's lip movements and the corresponding speech phonemes. These granular visual cues are directly linked to the Chain-of-Thought (CoT) text summary within the user interface, translating complex pixel-level anomalies into accessible narrative explanations. Ultimately, this transparent integration of visual and textual forensics ensures that end-users are not just given a binary classification, but are empowered with actionable evidence to confidently assess media authenticity in real time. This comprehensive approach significantly reduces the cognitive load required to identify sophisticated digital forgeries.

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Figure 5.7: Temporal Coherence Analysis graph. These forensic insights are then compiled into a comprehensive diagnostic report, which isolates the most suspicious frames. As seen in the generated report below, the system isolates Frame 15, successfully signaling background warping, mask misalignment, and blending errors around the subject and the football to the end user. By providing an explicit quantitative metric—specifically, an AI Confidence Score of 98.68% against a negligible Authenticity Score of 1.32%—the report establishes a definitive, statistically grounded baseline. This high degree of certainty is derived from the culmination of synchronous multi-modal feature extraction, translating complex network probabilities into a clear, unambiguous verdict.

Expanding upon the Temporal Coherence Analysis, the diagnostic framework meticulously scrutinizes inter-frame consistency to detect the unnatural timeline behavior typical of generative models. In the analyzed file, the peak anomaly identified at Frame 15 correlates with high-variance disruptions such as sudden texture shifts, dropped frames, or synthetic micro-expressions. Instead of merely scanning static images

independently, the temporal module evaluates transitional pixel states, catching the exact millisecond where the synthetic rendering engine fails to accurately calculate continuous kinematic motion.

Furthermore, the Spatial Artifact Localization section employs advanced Explainable AI (XAI) techniques to generate the superimposed forensic heatmap. The visual evidence presented demonstrates the network's mathematical gradients spiking

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aggressively directly over the football and the subject's hands. This localized "red zone" indicates severe blending errors or localized resolution mismatches precisely at the boundary where the human subject interacts with the physical object. By isolating these exact pixels, the system demystifies the forensic process and provides actionable, mathematically backed evidence of manipulation.

Figure 5.8: TruthSeeker AI Diagnostic Report. Finally, to maximize user comprehension and accessibility, the system features an interactive "Ask the AI Investigator" module driven by a local Large Language Model. This privacy-aware conversational interface allows users to directly interrogate the generated diagnostic report entirely on-device. When prompted to explain why the video was flagged, the local AI seamlessly synthesizes the mathematical forensic findings into an easy-to-understand natural language summary without exposing the user's conversational data to external cloud servers. Furthermore, users can

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initiate follow-up queries to drill down into specific frames or request clarification on the technical terminology used in the report. By bridging the gap between complex algorithmic outputs and human-readable insights, this conversational agent empowers non-technical users to make informed, confident decisions regarding media authenticity.

Figure 5.9: The local AI Investigator interface. In summary, the integration of the DAVID-XR1 framework and the local conversational AI agent transforms the TruthSeeker platform from a standard black box classifier into a transparent, user-centric forensic tool. By presenting complex mathematical anomalies through intuitive visual heatmaps, timeline graphs, and natural language summaries, the system effectively bridges the gap between advanced deep learning and practical media literacy. This ensures that end-users can confidently authenticate digital content while maintaining strict data privacy.

Skyra Android App Implementation

The Skyra forensic framework is meticulously engineered and deployed as a high performance mobile application, designated as the Skyra Forensic Suite, which is purpose-built to facilitate real-time, on-device deepfake detection directly at the edge. The underlying application is architecturally constructed using Kotlin and Jetpack Compose, a modern development stack selected specifically to provide a highly responsive, forensic-grade user interface capable of handling complex visual data and high-resolution video streams without significant latency. By integrating a specialized Multimodal Large Language Model (MLLM) into the mobile environment,

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the suite fundamentally transcends the restrictive boundaries of traditional binary classification systems. This advanced integration enables the system to not only detect anomalies but to actively identify human-perceivable visual artifacts and leverage them as grounded, spatio-temporal evidence, thereby providing both high-accuracy detection and comprehensive natural-language explanations that significantly enhance human interpretability.

As visualized within the intuitive mobile interface, the system employs a sophisticated and transparent reasoning-chain designed to surface even the most subtle and well-disguised deepfake signatures that might

otherwise evade standard detection methods. The application is functionally capable of identifying specific Forensic Markers situated within the video timeline, including nuanced anomalies such as Lighting Inconsistency and Shadow Discrepancy. These markers are analyzed by the Skyra AI Reasoning engine, which synthesizes detailed feedback for the investigator. For instance, the model can precisely articulate when a subject appears with direct, high-intensity sunlight that is mathematically and physically inconsistent with the softer, diffused ambient lighting observed on surrounding objects or vehicles in the background. Furthermore, the Ask Skyra interactive assistant empowers users to initiate a guided “description-inspection-review” protocol, facilitating granular forensic clarification by allowing the investigator to probe specific timestamps or suspected generative artifacts through a real-time conversational interface.

During the rigorous experimental validation of the mobile suite, the system was tasked with processing a complex AI-generated sequence—specifically a bartender interaction characterized by fluid and high-velocity object manipulation. The Skyra engine successfully isolated intrinsic, physics-violating artifacts, such as the physically impossible warping and deformation of a metal shaker, which began to contort as if made of a soft, malleable material during its movement. Additionally, the system identified the sudden, abnormal appearance of objects in the subject’s hand, such as dark metallic residue that had no logical or causal source within the preceding video frames. By localizing these specific anomalies using precise spatio-temporal tags and bounding boxes, the system returned a definitive forensic verdict grounded in the fundamental laws of physical reality, successfully avoiding the superficial visual cues and shortcut signals that frequently lead to false positives in less sophisticated models.

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Figure 5.10: Skyra Forensic Suite.

5.2.2 Audio Analysis

Parallel to the video pipeline’s general temporal anomaly detection, audio streams are subjected to rigorous verification. For deep inspection, the cloud modules bypass traditional Mel-spectrograms entirely, utilizing the Wav2Vec 2.0 Large architecture to directly process the raw 1-dimensional audio waveform. This allows the model’s convolutional feature extractor and 24-layer Transformer context network to capture microscopic acoustic phase anomalies, unnatural phonetic transitions, and synthetic vocal tract artifacts that are often lost or smoothed over during standard spectrogram conversion. This approach ensures that even hyper-realistic, deep-cloned audio tracks are accurately flagged.

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During the experimental evaluation, a known AI voice cloning sample (specifically, a synthetic recreation of the public figure Cristiano Ronaldo) was processed through the TruthSeeker deep audio scan module. The system successfully identified the underlying synthetic vocal tract artifacts, returning a definitive prediction of AI Generated (High Risk) with an AI Probability score of 91.79%.

Furthermore, the module’s Explainable AI (XAI) component—utilizing Integrated Gradients mapped the temporal coherence of the waveform to provide transparent forensic evidence. As shown in the diagnostic dashboard, the AI detected a localized breakdown in audio naturalness peaking at exactly 3.67 seconds. The extreme intensity of this mathematical anomaly gradient suggests a ‘hard glitch’ typical of digital splice points, unnatural breathing generation, or impossible shifts in vocal pitch common in state-of-the-art cloned audio.

Figure 5.11: Wav2Vec 2.0 classification results.

5.3 Quantitative Evaluation and Metrics

To quantitatively evaluate the performance of the deepfake detection models, standard classification metrics were utilized. The evaluation metrics include Accuracy, Precision, Recall, and F1-score.

• Accuracy: Measures the proportion of correctly classified samples among all 56 predictions.

Accuracy = $\frac{TP + TN}{TP + TN + FP + FN}$ (5.1)

• Precision: Measures the proportion of correctly predicted positive observations.

Precision = $\frac{TP}{TP + FP}$ (5.2)

• Recall: Measures the ability of the model to correctly identify manipulated samples.

Recall = $\frac{TP}{TP + FN}$ (5.3)

• F1-score: Represents the harmonic mean of Precision and Recall.

$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$ (5.4)

Where TP represents True Positives, TN represents True Negatives, FP represents False Positives, and FN represents False Negatives. Table 5.1 summarizes the performance of the individual analysis streams.

Table 5.1: Performance Metrics of TRUTH-SEEKER Modules Detection Module

Accuracy

Precision

Recall

F1-Score

Edge Model (Quick Scan) Frequency Analysis

Temporal Coherence Module Overall Hybrid System

88.5%

94.5%

95.2%

96.8%

87.2%

95.1%

94.8%

97.1%

89.1%

93.8%
96.0%
96.5%
88.1%
94.4%
95.4%
96.8%

5.4 System Performance Analysis

The system's hybrid architecture ensures a balance between processing speed and analytical depth. The lightweight edge model provides rapid predictions with an average inference time of approximately 20–30 milliseconds, making it highly suitable for real-time mobile applications. The cloud module, which performs the resource intensive multi-modal analysis (Fourier transforms and temporal sequence extraction), requires a higher processing time of approximately 1–3 seconds per video segment.

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However, this trade-off delivers significantly higher accuracy and generates the explainable AI visualizations (heatmaps and frequency graphs) required to build user trust.

Overall, the experimental results demonstrate that the TRUTH-SEEKER system effectively integrates localized mobile edge computing with advanced cloud-based deep learning to provide a robust, scalable, and highly accurate solution for combating deepfakes and misinformation.

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Chapter 6

Conclusion

The proposed TRUTH-SEEKER system demonstrates the feasibility of integrating deep learning, multi-modal analysis, and explainable artificial intelligence techniques to effectively detect deepfake content and verify textual information.

The developed system successfully combines edge-based detection, cloud-based analysis, multi-modal fusion, explainable AI (XAI), and fact verification within a unified framework. Experimental evaluation shows that the system achieves high detection accuracy while maintaining a balance between latency and computational efficiency through its hybrid architecture.

6.1 Project Summary

The primary objective of this project was to design and implement a hybrid system capable of detecting deepfake media and verifying the authenticity of textual information in real time.

The developed system integrates multiple modules including image and video analysis, audio processing, multi-modal fusion, explainable AI, and fact verification. For deepfake detection, the system uses both lightweight edge models and advanced cloud-based deep learning models to analyze media inputs.

The edge module performs fast initial screening using optimized models to provide immediate feedback. If the confidence level is insufficient, the input is forwarded to the cloud module for detailed analysis. The cloud system applies advanced techniques such as visual artifact detection, temporal inconsistency analysis, and audio anomaly

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detection.

The system further enhances transparency by incorporating explainable AI techniques such as heatmaps and saliency maps, which highlight manipulated regions and provide reasoning behind predictions.

For textual inputs, the system employs a Retrieval-Augmented Generation (RAG) approach to verify claims by retrieving relevant information from trusted sources and generating evidence-based explanations.

Experimental evaluation shows that the system achieves an overall detection accuracy of approximately 92–95%, depending on the input modality and dataset. The hybrid design ensures low latency for real-time detection while maintaining high accuracy through cloud-based analysis.

The integration of multi-modal detection, explainable AI, and fact verification enables the system to provide a comprehensive solution for combating deepfakes and misinformation.

6.2 Key Contributions

The major contributions of this project can be summarized as follows:

- Development of a hybrid deepfake detection system combining edge computing and cloud-based analysis.
- Implementation of a multi-modal detection framework integrating visual, audio, and temporal features.
- Integration of Explainable AI (XAI) techniques to improve transparency and user trust.
- Design of a fact verification module using Retrieval-Augmented Generation (RAG) for detecting misinformation..
- Development of a highly scalable and efficient system optimized for real-time analysis, meticulously balancing rigorous forensic accuracy with low inference latency, and comprehensively benchmarked against state-of-the-art datasets to ensure operational resilience.

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These contributions demonstrate the potential of combining artificial intelligence and multi-modal data analysis to build reliable systems for detecting deepfakes and ensuring information authenticity

6.3 Limitations

Although the proposed system demonstrates strong performance, several limitations remain.

The effectiveness of deepfake detection depends heavily on the quality and diversity of the training dataset. The system may face challenges when encountering highly sophisticated or previously unseen deepfake generation techniques.

Edge-based models, while efficient, may not capture subtle manipulations due to their lightweight design. This makes cloud-based processing necessary for high confidence predictions.

Additionally, cloud processing introduces latency and requires stable network connectivity, which may affect real-time performance in certain scenarios. The fact verification module relies on the availability and reliability

of external data sources. In cases where accurate or updated information is not available, verification results may be limited.

6.4 Future Work

Several improvements can be explored in future work to enhance the system's performance and capabilities. One important direction is expanding the dataset to include newer and more diverse deepfake generation techniques, which would improve generalization and detection accuracy. Future work may also involve exploring advanced deep learning architectures such as transformer-based models for improved multi-modal fusion and temporal analysis. Another potential improvement is enhancing the explainability module by incorporating more advanced visualization techniques and user-friendly explanations. The system can also be optimized for deployment on mobile and edge devices using model compression techniques such as quantization and pruning. Furthermore, integrating real-time web-based fact-checking

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APIs and continuously updated knowledge bases can improve the reliability and accuracy of the verification module. Finally, the system can be extended to detect other forms of misinformation such as manipulated text, synthetic media, and AI-generated content across social media platforms.

Overall, the proposed TRUTH-SEEKER system provides a strong foundation for future research in deepfake detection and misinformation control, highlighting the potential of artificial intelligence in ensuring digital media authenticity and trust.

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